

Report: Network Exclusion and State Collapse: From Maritime Isolation to Technological Access Denial in the Long Run of History

Thank you for giving me the chance to read and comment on this paper. First and foremost, this is an ambitious paper, which is laudable. It seeks to argue that one of the main explanations for state collapse could be technological exclusion, resulting not from policy choices, but insurmountable structural features like geography. Given the importance of technology, I think there's some merit here. However, whether technological exclusion is going on feels rather assumed than shown, which strikes me as a major issue. Additionally, I think the paper has several empirical problems around the author's classifications.

Classification Issues

Key to this paper's success is its classification of polities that were excluded from technical networks. The author identifies seven main candidates here, ranked on a continuum of "moderate" to "strong."

I think some of these classification choices are extremely questionable. For example, one of the candidates is the Timurid Empire, because it is "landlocked" and connected via the Silk Road only. But this isn't true – the Timurid Empire *wasn't* landlocked. There are several water access points, mainly around the Persian Gulf. Perhaps the argument is that these weren't the dominant trading networks at the time – but claiming the Timurid empire is landlocked is just not right. Does the author mean certain economic or political centers of power were disconnected from the Mediterranean? That would be a more defensible position, but the author never clarifies. This is a major issue with this classification.

A bigger issue may lie with the Han Empire. The Han Empire is said to be excluded because it was excluded from the Mediterranean network (true) but that the Silk Road provided "limited" overland contact. "Limited" feels like it is doing an immense amount of unjustified and unexplained lifting here – from what I recall, the Silk Road was a pretty big deal! This is also confounded by the fact that the Silk Road is *clearly identified by the author themselves* as a "dominant technological platform." To quote the paper:

"Each era has its own dominant technological platform – the infrastructure through which frontier knowledge diffuses across states. In antiquity, it was the Mediterranean trade routes and Silk Road caravans." – pp. 31 – 32.

So... does the Silk Road matter, or does it not? If it doesn't matter, then the author is misrepresenting this in their own paper. If it *does* matter, then I don't see a strong argument at all for counting Han China as technologically excluded. The same goes for the Timurid Empire, which, again, was not landlocked and also benefited from the Silk Road. One could also make a very strong argument that the author is underemphasizing the importance of the Dnieper river trade route for the Kievan Rus, and connecting them to the technological center. That trade route was immensely

important in connecting them to the Byzantine Empire, which was a dominant player technologically at the time.

So, I think there are severe flaws with at least two, maybe three, of the seven classifications the author has made. That is a huge deal when the sample size is as small as it is. I am not sure if I buy the classifications used, which means I don't think I buy the paper's empirical contribution.

I think the Kievan Rus story also complicates the narrative of technological exclusion. When they were conquered, it certainly was due to tactical and technological superiority, but wouldn't the Golden Horde under Batu Khan count as *more* technologically excluded in the author's own estimation? They would have been even further away from the technological hub of the Mediterranean than the nation they had conquered! Unless of course the argument is that they benefited from controlling parts of the Silk Road, which brings the question of what the correct technological locus was to the forefront once again.

Concerns Applying the Criteria

The classification scheme, as described above, is loose and sometimes appears internally contradictory. This raises further concerns about the fact that all the seven technologically excluded polities were eventually conquered. Perhaps that would be the case if a more objective and clear classification structure were established, but there isn't one. That makes me concerned that the author's discretion may be playing a larger role in the selection of these cases. Fixing the classification scheme and being much more rigorous about it would help to obviate this concern, but this is not done in the current version of the paper.

Also... it is not surprising that moving the seven polities the author has selected from the open category to the closed category strengthens the significance of closure on being conquered. This is because all seven polities the author moves were indeed conquered – we know this already. So I don't think this actually strengthens our inference that technological closure makes conquest more likely. It's a mathematical certainty that moving conquered polities that were originally open to the closed category will reduce the conquest percentage of the former and increase the conquest percentage of the latter. I don't think that seeing significance increase as the polities are moved to the closed group indicates they belong in the closed group at all – it's just what will happen automatically as you move conquered polities from one group to the other. I think the author needs a coding strategy/guide that convincingly shows their classification choices were truly done blind.

Mechanism Gaps

The paper's proposed mechanism feels plausible. The argument is that being cut off from the technological heartbeat of the era will accelerate or lead to state collapse. I think the author is successful in a few places (see above) at identifying some polities that were separated from others via structural or geographic reasons. But there's no work done to show that there was a disruption of technological transmission – again, it's just sort of assumed. Maybe it's a lack of technological access that's the issue, but maybe it's something different. Joseon Korea kind of complicates this. Joseon

Korea was closed as a policy choice, and still kept access to sources of technological knowledge, but they still were annexed. I think the paper would be well-served by providing some clear examples of technological disparities that existed, and some plausible economically informed arguments as to why those technologies could not diffuse because of structural reasons. I just don't think it's substantiated well enough in this paper.

The Political Economy Story

The political economy connection feels tenuous at best. I don't know what the selectorate theory stuff does at the beginning for the paper, and I do tend to like that theory. But it doesn't seem to play a role later on in the paper's story – the tale, as far as I can tell, is that exclusion leads to worse economic performance and declining state capacity, which in turn makes the country easier to conquer. However, there's no work done to demonstrate that the polities which were technologically excluded suffered more economically or faced reduced levels of state capacity – again, it's just kind of assumed. So the meat of the author's political economy story seems to be missing, which makes me concerned about this paper's fit in *Public Choice*.

Conclusion

Again, this is an interesting and ambitious paper. I truly do commend the author for trying to pull something off like this. But the current version of the paper just isn't successful, and I think a successful version of this paper requires a complete re-write and much more work. The classification issues have to be fixed. There needs to be much more historical work showing declining economic performance and state capacity in the technologically excluded polities relative to those that were excluded via a policy choice. More careful work needs to be undertaken showing that technology really was where the action was, and not something else. Because of this, I am recommending the paper be rejected, but I do encourage the authors to continue along this path of inquiry. It's a big question with an appealing mechanism behind it.